

Dataset Overview

Characteristic	Value
Source	GSE300475 (DFCI 16-466 trial)
Patients	4
Longitudinal samples	11
Total cells	100,067
Genes profiled	~20,000
Cells with productive TCR	≈74%
Feature families	4 (29–429 features)
Cross-validation	Leave-one-patient-out

TCR-Only Negative Control (Cell-Level)

Feature Set	Accuracy	Precision	Recall	AUC
Comprehensive (multimodal)	0.928	0.878	0.936	0.978
Seq. Structure (multimodal)	0.927	0.877	0.934	0.978
TCR-Only	0.675	0.532	1.000	0.484

TCR-only encodings produce a trivial high-recall classifier (recall = 1.0, AUC = 0.484), consistent with predicting all cells as responders. This negative-control result demonstrates that TCR-derived features alone do not drive cell-level discrimination in this cohort.

Limitations

- Small cohort (4 patients, 11 samples)
- Retrospective dataset - no prospective validation
- Cell-level metrics may not directly translate to patient-level generalizability
- External validation in independent cohorts required